

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Ethical Hacking [NPT_4125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Implement how TCP three-way handshake behavior can be exploited during reconnaissance. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Apply how subnetting assists in isolating network segments to improve security and performance. (5)
Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Examine the security implications of routing protocols such as RIP and OSPF in dynamic networks. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Implement the role of Google Dorks and archive.org in passive information gathering. (5)
Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Explain how MAC spoofing affects ARP tables and network integrity. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Understand"
B) Analyze how symmetric and asymmetric encryption schemes differ in performance and security. (5)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 4) Analyze how cryptographic hash functions ensure data integrity and authenticity in message transmission. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Examine how steganography conceals information compared to encryption, with examples. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 5) Analyze how packet sniffing via Wireshark can expose credentials and how encryption mitigates the risk. (5)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
B) Evaluate lessons learned from real-world case studies of DDoS attacks and mitigation techniques. (5)

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Event Name	Subject
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Hacking

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)